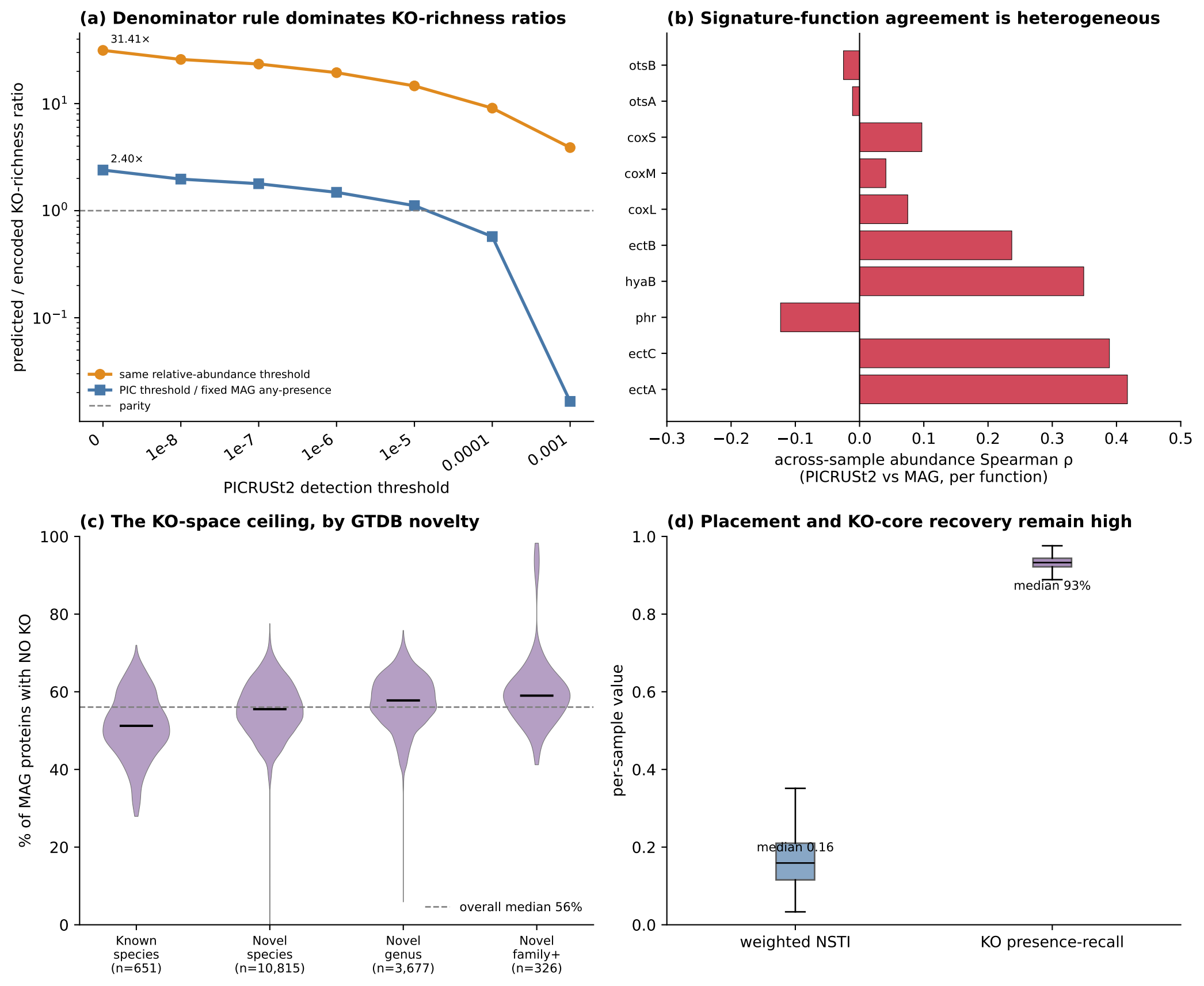
(a) Denominator rule dominates KO-richness ratios
31.41×
2.40×
predicted / encoded KO-richness ratio
same relative-abundance threshold
PIC threshold / fixed MAG any-presence
parity
0
1e-8
1e-7
1e-6
1e-5
0.0001
0.001
PICRUSt2 detection threshold
(b) Signature-function agreement is heterogeneous
otsB
otsA
coxS
coxM
coxL
ectB
hyaB
phr
ectC
ectA
across-sample abundance Spearman ρ
(PICRUSt2 vs MAG, per function)
(c) The KO-space ceiling, by GTDB novelty
% of MAG proteins with NO KO
overall median 56%
Known species (n=651)
Novel species (n=10,815)
Novel genus (n=3,677)
Novel family+ (n=326)
(d) Placement and KO-core recovery remain high
per-sample value
median 93%
median 0.16
weighted NSTI
KO presence-recall